Data Structure Design: Entity Relational Schema & Data Dictionaries

Phase 3: Database & DataFrame Specifications

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1. Relational Schema & Attribute Dictionary

The relational data model consists of a normalized star schema centered on individual athlete appearances joined with national aggregated medal ledgers.

Table 3.1: Athletes Schema Data Dictionary (athletes.csv)

Field Name	Data Type	Constraint	Description
id	Integer	NOT NULL	Unique athlete identifier
name	String	NOT NULL	Full athlete legal name
sex	String (1)	M / F	Biological sex of athlete
age	Float	Nullable	Age of athlete at time of competition
height	Float	Nullable	Height in centimeters
weight	Float	Nullable	Weight in kilograms
noc	String (3)	NOT NULL	Three-letter National Olympic Committee code
country	String	NOT NULL	Standardized sovereign country name
games	String	NOT NULL	Olympic Games edition label (e.g., '2016 Summer')
year	Integer	1980 - 2016	Four-digit calendar year of competition
season	String	Summer / Winter	Olympic seasonal classification
city	String	NOT NULL	Host city of the Olympic Games
sport	String	NOT NULL	Governing Olympic sports discipline
event	String	NOT NULL	Specific competitive event description
medal	String	Domain Validated	Gold, Silver, Bronze, or No Medal
medal_value	Integer	0, 1, 2, 3	Weighted score equivalent (Gold=3, Silver=2, Bronze=1, None=0)
is_medal	Integer	0 / 1	Binary flag indicating whether athlete reached the podium